TIMILEHIN OLAPADE

Independent AI researcher | Adaptive decision systems | Causal learning | Multiagent reliability

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RESEARCH PROFILE

Independent researcher and founder investigating **learning and decision making in adaptive environments**, especially where deployed models influence the processes they attempt to predict. My work combines operational AI systems, causal identification, regime conditional expert selection, quantitative simulations and decision auditability. Current questions include collective failure in interacting autonomous agents, uncertainty aware control, predictive degradation under distribution shift, and the distinction between apparent performance improvements and genuinely better inference. I use financial markets as a testbed for general questions about AI reliability.

SYSTEMS ARCHITECTURE | EXPERIMENTAL VALIDATION

Learning and state inference	Regime-conditioned pipelines integrating probabilistic state estimation, statistical and neural prediction, feature research and adaptive model selection.
Evidence-gated autonomy	Decoupled model orchestration, structured decision checks, uncertainty-aware abstention, exposure limits, circuit breakers and execution interfaces.
Research to operational feedback	A private deployed decision architecture informed reproducible audit methods, controlled market-feedback experiments and independently testable research components.

Internal prototype testing (self-reported): a return above 20% over one quarter. A separate historical ATRX performance summary reported a Sharpe ratio of approximately 2.1. These figures have **not been independently audited** and have not been established as measurements from the same test. Test dates, cost treatment, benchmark and return series require reconciliation; neither number is presented as validated persistent alpha.

RESEARCH MANUSCRIPTS | SINGLE AUTHOR

01 Adaptive Participation and Market Feedback: A Deployment Audit and Reproducible Simulation Study

2026 | Submitted to AISTATS 2027; under review, decision pending | NOT an accepted paper

Empirically audits a deployed adaptive decision architecture, reconstructing observability across **28,165 logged events** and comparing **four model-output variants on 5,998 shared inputs**. Complements the audit with **1,312 controlled synthetic episodes** to examine participation and shock interactions. Explicitly separates observed model agreement, predictive validity and conditional simulation results.

Research manuscript (PDF) | Replication code

02 Market Transitions under AI Financial Models: Alpha Decay, Instability, and Evidence-Gated Neuro-Symbolic Adaptation

31 October 2026 | Public research working paper; not peer reviewed

Investigates adaptive model participation and systemic market feedback through **structural estimation diagnostics** and an **interacting population of 24 simulated agents**, spanning **256 parameter configurations and 64 replications per configuration**. Examines crowding, liquidity, leverage, adjustment speed and decision eligibility; distinguishes lower losses from better forecasts and reports sensitivity to model assumptions.

Manuscript, code and results | Reproduction instructions

03 Agentic Causal Macro Intelligence: An Identification-Aware Framework for Regime-Conditional Expert Routing

31 October 2026 | Revised public working paper; revision of September 2026 Zenodo / ResearchGate preprint

Proposes an **evidence-constrained mixture of experts** architecture connecting regime uncertainty, explicit causal hypotheses, expert eligibility and abstention. Public diagnostic code generates **800 synthetic datasets and 4,000 method rows** to investigate identification and model specification failures. Shared diagnostics with paper 02 are disclosed and not presented as independent evidence.

Revised manuscript and code | ResearchGate preprint | Zenodo DOI

Related derivative: a shorter Agenthon / workshop formatted manuscript is available in the paper 01 repository; it is not counted as an independent peer reviewed research publication.

AI SYSTEMS ARCHITECTURE AND RESEARCH ENGINEERING

Adaptive model orchestration and risk constrained decision systems. Architected a modular research-to-deployment stack joining streaming market data, feature computation, probabilistic regime inference, supervised and sequential predictors, contextual decision validation, position and exposure controls, and execution interfaces. Designed model and risk components to be replaceable and independently testable. The public repository is a reference implementation; private operational systems and model artifacts are not included.

Learning under endogenous feedback. Developed audit and simulation methods for studying environments that change after automated systems intervene. The design distinguishes predictive error from changes in realized outcomes caused by market participation, leverage and risk gating. Theoretical market mechanisms are tested in controlled synthetic environments rather than asserted as demonstrated causal properties of live markets.

Causal evidence and expert routing. Developed an identification aware research framework using explicit macroeconomic transmission hypotheses and dynamic model selection. The principal engineering contribution is a **decision evidence contract**: model proposals must be interpreted alongside regime validity, evidence eligibility, uncertainty, exposure constraints and the option to abstain.

Reproducible evaluation infrastructure. Published configuration controlled simulation suites, deterministic test runners, sensitivity analyses, result manifests and reusable code. Emphasizes chronological separation, baseline comparisons, ablations, paired contrasts, provenance and honest reporting of unobserved outcomes.

PUBLIC SOFTWARE AND IMPLEMENTATION EVIDENCE

- **Adaptive AI Market Transitions**: Operational audit procedures, controlled participation simulations, contract tests, reproducible manuscript.
- **Market Transitions under AI Financial Models**: Market ecology simulations, structural estimation, sensitivity studies and evidence gated routing.
- **Identification Aware Expert Routing**: Causal identification diagnostics, expert routing framework and repeatable benchmark scripts.
- **ATRX Public Reference Architecture**: Multiasset inference and risk system reference: market data, feature generation, regime models, expert validation, risk controls and execution adapters.
- **Quantitative Research Engine**: Experiment pipeline: feature engineering, XGBoost/LSTM/CNN, hidden state regime analysis, temporal validation, backtesting and robustness analysis.
- **FX Volatility Predictor**: Time series forecasting and time ordered evaluation for volatility research.
- **Adaptive FX Scalper**: Research code for regime dependent strategy selection and volatility sensitive risk management.

FOUNDER AND RESEARCH EXPERIENCE

Haldane Technologies Inc. (Delaware, United States) / ATRX Intelligence | Founder and research lead / principal engineering contributor | 2025 to present

- Lead independent research and systems engineering spanning adaptive AI architectures, quantitative modeling, uncertainty aware decisions, market feedback and evaluation infrastructure.
- Designed an operational AI decision and execution architecture that subsequently enabled systematic audit of model outputs, execution responses and deployment observability. This experience informs research questions. Preliminary, self-reported pilot performance is separately disclosed and has not been independently validated.
- Maintain separation between proprietary operational infrastructure and open research packages, and between demonstrated simulation findings and untested next stage architectures.

ACTIVE RESEARCH AGENDA | NOT YET COMPLETED

Financial and economic world models; latent state representation learning; calibrated state transition and decision models; model degradation and alpha decay; decentralized agent monitoring; coordinated failure detection; causal hypotheses under regime shifts; neuro symbolic constraint validation; human oversight and safe abstention. These concepts are **research directions**, not claims of a trained foundation world model or verified safety intervention.

APPLIED ANALYTICS AND INSTITUTIONAL LEADERSHIP

Union Bank of Nigeria PLC, Lagos | Tax Management Reporting Team Lead | June 2023 to present

- Lead audit ready enterprise reporting, statutory and management reconciliations, tax controls and evidence traceability across business, financial reporting and technology functions.
- Apply Oracle SQL, Python and Power BI to large scale enterprise data reconciliation, anomaly investigation, operational decision support and regulatory responses.
- Contributed analytical work to reported cumulative tax savings above **NGN 8 billion**; coordinated regulatory electronic invoicing work and VAT automation associated with approximately **40% faster processing** and **95% workflow accuracy**. These are internal reported outcomes, not independently audited impact estimates.

EDUCATION

WorldQuant University | MSc Financial Engineering | In progress; expected August 2027

Graduate training in financial econometrics, stochastic processes and modeling, portfolio and risk optimization, computational finance and machine learning. **Foundations of Financial Engineering**, eight graduate credits completed September 2026.

Adeleke University | BSc Business Administration | 2021

Best Graduating Student, Business Administration; cumulative GPA 4.0/5.0.

METHODS, LANGUAGES AND RESEARCH PRACTICE

Software and data: Python, SQL / Oracle, C and C++ (algorithmic programming), Git, Linux workflows, Power BI, MetaTrader 5 interfaces, NumPy, pandas, SciPy, scikit-learn, XGBoost.

Quantitative modeling: stochastic simulation, hidden Markov and mixture regime models, time series forecasting, recurrent and convolutional networks, Monte Carlo experiments, stress testing, constrained decision rules, causal graphical hypotheses and mixture of experts model selection.

Research standards: replicable experiment configurations, chronological holdouts, comparisons against simple baselines, sensitivity analysis, ablation tests, matched experimental contrasts, evidence provenance and explicit separation of observed and simulated outcomes.

AWARDS, CREDENTIALS AND PROFESSIONAL DEVELOPMENT

Best Graduating Student, Adeleke University (2021); WorldQuant University Foundations of Financial Engineering (2026, eight graduate credits); HackerRank SQL Advanced and Problem Solving Intermediate assessments (2025); MathWorks Reinforcement Learning Onramp completion (2025); Microsoft Develop a Vision Enabled Generative AI Application course credential (2025). ICAN ACA professional qualification in progress.

RELEVANCE TO THE FORESIGHT INSTITUTE GRANT

World Models for Safe Coordination in Adaptive Multiagent Systems (proposed project). Extend existing research on model induced feedback and evidence constrained decisions into an openly reproducible safety benchmark. Compare baseline autonomous agents with learned world state models and independent verifier components in synthetic market and shared resource environments. Evaluate correlated failures, risk detection quality, time to intervention, unnecessary intervention rates, and effects on productive cooperation. Research outputs would be open source and separate from proprietary ATRX systems.

RESEARCH INTEGRITY AND ACCESS

The AISTATS manuscript is **under review, not accepted**. The other listed papers are public working papers or preprints, **not peer reviewed journal articles**. Public code supports synthetic replication, while restricted operational logs cannot be fully reproduced externally. The September ResearchGate / Zenodo preprint precedes the narrower October identification aware revision; its broad implementation and performance claims should not be treated as independently verified. Preliminary testing returns and a historical Sharpe figure are self-reported, not independently audited or reconciled to one verified performance series. No robust causal discovery or completed financial world model is claimed. References available on request.

Public research identity: GitHub | Scholar | ResearchGate paper | Zenodo | ORCID